

DPP-9

- Q.1 The value of 'a' for which the lines $\frac{x-2}{1} = \frac{y-9}{2} = \frac{z-13}{3}$ and $\frac{x-a}{-1} = \frac{y-7}{2} = \frac{z+2}{-3}$ intersect, is
(A) -5 (B) -2 (C) 5 (D) -3
- Q.2 Given A(1, -1, 0); B(3, 1, 2); C(2, -2, 4) and D(-1, 1, -1) which of the following points neither lie on AB nor on CD?
(A) (2, 2, 4) (B) (2, -2, 4) (C) (2, 0, 1) (D) (0, -2, -1)
- Q.3 For the line $\frac{x-1}{1} = \frac{y-2}{2} = \frac{z-3}{3}$, which one of the following is incorrect?
(A) it lies in the plane $x - 2y + z = 0$ (B) it is same as line $\frac{x}{1} = \frac{y}{2} = \frac{z}{3}$
(C) it passes through (2, 3, 5) (D) it is parallel to the plane $x - 2y + z - 6 = 0$
- Q.4 Given planes
 $P_1 : cy + bz = x$
 $P_2 : az + cx = y$
 $P_3 : bx + ay = z$
 P_1, P_2 and P_3 pass through one line, if
(A) $a^2 + b^2 + c^2 = ab + bc + ca$ (B) $a^2 + b^2 + c^2 + 2abc = 1$
(C) $a^2 + b^2 + c^2 = 1$ (D) $a^2 + b^2 + c^2 + 2ab + 2bc + 2ca + 2abc = 1$
- Q.5 The line $\frac{x-x_1}{0} = \frac{y-y_1}{1} = \frac{z-z_1}{2}$ is
(A) parallel to x-axis (B) perpendicular to x-axis
(C) perpendicular to YOZ plane (D) parallel to y-axis
- Q.6 The lines $\frac{x-2}{1} = \frac{y-3}{1} = \frac{z-4}{-k}$ and $\frac{x-1}{k} = \frac{y-4}{2} = \frac{z-5}{1}$ are coplanar if
(A) $k = 0$ or -1 (B) $k = 1$ or -1 (C) $k = 0$ or -3 (D) $k = 3$ or -3
- Q.7 The line $\frac{x-2}{3} = \frac{y+1}{2} = \frac{z-1}{-1}$ intersects the curve $xy = c^2$, in xy plane if c is equal to
(A) ± 1 (B) $\pm 1/3$ (C) $\pm \sqrt{5}$ (D) none
- Q.8 The line which contains all points (x, y, z) which are of the form $(x, y, z) = (2, -2, 5) + \lambda(1, -3, 2)$ intersects the plane $2x - 3y + 4z = 163$ at P and intersects the YZ plane at Q. If the distance PQ is $a\sqrt{b}$ where $a, b \in \mathbb{N}$ and $a > 3$ then $(a + b)$ equals
(A) 23 (B) 95 (C) 27 (D) none
- Q.9 Let L_1 be the line $\vec{r}_1 = 2\hat{i} + \hat{j} - \hat{k} + \lambda(\hat{i} + 2\hat{k})$ and let L_2 be the line $\vec{r}_2 = 3\hat{i} + \hat{j} + \mu(\hat{i} + \hat{j} - \hat{k})$.
Let Π be the plane which contains the line L_1 and is parallel to L_2 . The distance of the plane Π from the origin is
(A) $1/7$ (B) $\sqrt{2/7}$ (C) $\sqrt{6}$ (D) none
- Q.10 The value of m for which straight line $3x - 2y + z + 3 = 0 = 4x - 3y + 4z + 1$ is parallel to the plane $2x - y + mz - 2 = 0$ is
(A) -2 (B) 8 (C) -18 (D) 11

- Q.11 A straight line is given by $\vec{r} = (1+t)\hat{i} + 3t\hat{j} + (1-t)\hat{k}$ where $t \in \mathbb{R}$. If this line lies in the plane $x + y + cz = d$ then the value of $(c + d)$ is
 (A) -1 (B) 1 (C) 7 (D) 9
- Q.12 The distance of the point $(-1, -5, -10)$ from the point of intersection of the line $\frac{x-2}{2} = \frac{y+1}{4} = \frac{z-2}{12}$ and the plane $x - y + z = 5$ is
 (A) $2\sqrt{11}$ (B) $\sqrt{126}$ (C) 13 (D) 14
- Q.13 $P(\vec{p})$ and $Q(\vec{q})$ are the position vectors of two fixed points and $R(\vec{r})$ is the position vector of a variable point. If R moves such that $(\vec{r} - \vec{p}) \times (\vec{r} - \vec{q}) = 0$ then the locus of R is
 (A) a plane containing the origin 'O' and parallel to two non collinear vectors $\vec{OP}$ and $\vec{OQ}$
 (B) the surface of a sphere described on PQ as its diameter.
 (C) a line passing through the points P and Q
 (D) a set of lines parallel to the line PQ .

MATCH THE COLUMN:

- Q.14 Consider the following four pairs of lines in **column-I** and match them with one or more entries in **column-II**.

Column-I	Column-II
(A) $L_1 : x = 1 + t, y = t, z = 2 - 5t$ $L_2 : \vec{r} = (2, 1, -3) + \lambda(2, 2, -10)$	(P) non coplanar lines
(B) $L_1 : \frac{x-1}{2} = \frac{y-3}{2} = \frac{z-2}{-1}$ $L_2 : \frac{x-2}{1} = \frac{y-6}{-1} = \frac{z+2}{3}$	(Q) lines lie in a unique plane
(C) $L_1 : x = -6t, y = 1 + 9t, z = -3t$ $L_2 : x = 1 + 2s, y = 4 - 3s, z = s$	(R) infinite planes containing both the lines
(D) $L_1 : \frac{x}{1} = \frac{y-1}{2} = \frac{z-2}{3}$ $L_2 : \frac{x-3}{-4} = \frac{y-2}{-3} = \frac{z-1}{2}$	(S) lines are not intersecting

- Q.15 $P(0, 3, -2)$; $Q(3, 7, -1)$ and $R(1, -3, -1)$ are 3 given points. Let L_1 be the line passing through P and Q and L_2 be the line through R and parallel to the vector $\vec{V} = \hat{i} + \hat{k}$.

Column-I	Column-II
(A) perpendicular distance of P from L_2	(P) $7\sqrt{3}$
(B) shortest distance between L_1 and L_2	(Q) 2
(C) area of the triangle PQR	(R) 6
(D) distance from $(0, 0, 0)$ to the plane PQR	(S) $\frac{19}{\sqrt{147}}$